COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structure and AlgorithmS	**SECTION:**	CPE1S2
DATE PERFORMED:	14/10/2026	**DATE SUBMITTED:**	14/10/2026
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Hands-on Activity 12.1 Algorithmic Strategies

1. PROCEDURE OUTPUT

ILO.A

Strategy	Algorithm	Analysis
Recursion	Binary search	Splits the stored elements into half then leaving only the relevant half then repeatedly breaks it down each into smaller portions until the target is found but unlike the divide and conquer it can stop at a smaller cutoff subportion the moment the element is found early
Brute force	Linear search	Exhaustively looks through every single candidate solution in the search space sequentially without optimization. For instance, iterating through every element in an array one by one until a match is found.
Back tracking	Depth first search	Builds candidate solutions incrementally along a state-space tree. When a path violates constraints, it backtracks up to the previous checkpoint to explore alternative branches.
Greedy	Selection sort	After every iteration the algorithm looks into the unsorted portion of the array then looks for the best candidate in the element one by one which is the minimum or maximum then swaps them
Divide and conquer	Merge sort	Merging sort splits the list and or

		array into smaller sections until there is only one left for each interval, sorting them in order and finally combining them all together in order.

Table 12-1. Algorithmic Strategies and Examples

Screenshot	Analysis


`#include <iostream>`
`#include <vector>`
`#include <algorithm>`
 	
`std::vector<int> memo;`
 	
`int getMinStepsMemo(int n) {`
`  if (n == 1) return 0;`
`  if (memo[n] != -1) return memo[n];`
 	
`  int r = 1 + getMinStepsMemo(n - 1);`
 	
`  if (n % 2 == 0) {`
`  r = std::min(r, 1 + getMinStepsMemo(n / 2));`
`  }`
`  if (n % 3 == 0) {`
`  r = std::min(r, 1 + getMinStepsMemo(n / 3));`
`  }`
 	
`  memo[n] = r;`
`  return r;`
`}`
 	
`int main() {`
`  int testCases[] = {1, 4, 7, 10};`
`  std::cout << "--- Memoization Top-Down Results ---" << std::endl;`
`  for (int n : testCases) {`
`  memo.assign(n + 1, -1);`
`  std::cout << "Min steps for n = " << n << " : " << getMinStepsMemo(n) << std::endl;`
`  }`
`  return 0;`
`}`
 	
`--- Memoization Top-Down Results ---`
`Min steps for n = 1 : 0`
`Min steps for n = 4 : 2`
`Min steps for n = 7 : 3`
`Min steps for n = 10 : 3`
	To solve the problem of Minimum Steps to One the Top-Down Memoization approach starts at n and recursively computes the answer for the steps taken: -1, x2, x3. The repeated computation down multiple branches of a recursion is eliminated by keeping track of the number of steps already done using a lookup table memo, which is initially set to -1 then after the

Table 12-2. Memoization Implementation

Screenshot	Analysis

Table 12-3. Bottom-Up Dynamic Programming Implementation

note : procedures are both groupings same for the supplementary except for the analysis of sup

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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